

<page_start no = 1>

<sol_start id=1>

1. $(3) 3$

<sol_end>

<sol_start id=2>

2. $\tan 60^\circ$

<sol_end>

<sol_start id=3>

3. 100°

<sol_end>

<sol_start id=4>

4. 10 cm

<sol_end>

<sol_start id=5>

5. $a = 0$

<sol_end>

<sol_start id=6>

6. Four decimals places

<sol_end>

<sol_start id=7>

7. 3 and 1

<sol_end>

<sol_start id=8>

8. $(0, -1)$

<sol_end>

<sol_start id=9>

9. No solution

<sol_end>

<sol_start id=10>

10. $k = \frac{9}{2}$

<sol_end>

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<page_start no = 2>

<sol_start id=11>

11. $k \neq 3$

<sol_end>

<sol_start id=12>

12. 8

<sol_end>

<sol_start id=13>

13. similar

<sol_end>

<sol_start id=14>

14. $x^3y^4z^4$

<sol_end>

<sol_start id=15>

15. $u^2 - u + 12$

<sol_end>

<sol_start id=16>

16. Intersecting coincident

<sol_end>

<sol_start id=17>

17. Parallel

<sol_end>

<sol_start id=18>

18. 16 : 81

<sol_end>

<sol_start id=19>

19. Both AUR are correct but R is not the correct explanation of A.

<sol_end>

<sol_start id=20>

20. A is false but R is true.

<sol_end>

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<page_start no = 3>

<sol_start id=21>

21. $6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$

$$\begin{array}{r|l} 5 & 1 \\ \hline 5 & 4 \\ \hline 2 & 5 \\ \hline 9 & 15 \end{array}$$

$$= 5 \times 5 \times 29$$

Since it is the product of the prime it is a composite number.

<sol_end>

<sol_start id=22>

22. $4x + 6y = 7$, $kx + 18y = 21$

$$\frac{4}{k} = \frac{6}{18} \times \frac{7}{21}$$

$$6k = 72$$

$$k = 12$$

k should be equal to 12.

<sol_end>

<sol_start id=23>

23. $(0, 0), (-6, 8)$

$$\sqrt{x^2 + y^2} = \sqrt{36 + 64}$$

$$= 10$$

<sol_end>

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<page_start no = 4>

<sol_start id=24>

$$\begin{aligned} 24. \quad \sin 3A &= \cos A - 10 \\ &= \cos 90 - 3A = \cos A - 10 \\ \sin \theta &= \cos 90^\circ - 0 \\ 90 - 3A &= A - 10 \\ 90 + 10 &= A + 3A \\ 4A &= 100 \\ A &= \frac{100}{4} = 25^\circ \end{aligned}$$

<sol_end>

<sol_start id=25>

$$\begin{aligned} 25. \quad \angle AOB &= \angle DOC && [\text{Vertical angles}] \\ \angle OAB &= \angle ODC && [\text{Alternate angles}] \\ \angle ABO &= \angle DCO && [,, \quad ,,] \\ \text{from } A - A - A \\ \triangle AOB &\sim \triangle DOC \end{aligned}$$

<sol_end>

<sol_start id=26>

$$\begin{aligned} 26. \quad \text{W.K.T} \\ \cos(90 - \theta) &= \sin \theta \\ \sin(90 - \theta) &= \cos \theta \\ \tan(90 - \theta) &= \cot \theta \\ &= \sin(50 + \theta) = \cos(40 - \theta) = \tan 1 \times \tan 10 \\ \tan 20 \times \tan 70 \times \tan 80 \times \tan 89 \\ &= \cos(40 - \theta) - \cos(40 - \theta) + \cot 89^\circ \times \cos 80^\circ \times \cot 70^\circ \times \tan 70^\circ \times \tan 80^\circ \times \tan 89^\circ \\ &= \cot 89 \cdot \cot 80 \cdot \cot 70 \cdot \tan 70 \cdot \tan 80 \cdot \tan 89 \\ &= \cot \theta = \frac{1}{\tan} \end{aligned}$$

$$\text{so } \frac{\cancel{\tan 70} \cdot \cancel{\tan 80} \cdot \cancel{\tan 89}}{\cancel{\tan 70} \cdot \cancel{\tan 80} \cdot \cancel{\tan 89}} = 1$$

Hence proved.

<sol_end>

<sol_start id=27>

27. It states that if a line is parallel to the side of a triangle which intersects the other side into two distinct points then the line divides those sides of the triangle in proportion.

Let ABC be a triangle

To prove $\frac{AD}{DB} = \frac{AF}{EC}$

Join BE, CD

Draw $EF \perp AB, DG \perp CA$

Since $EF \perp AB, DG \perp CA$

EF is the height of triangle ADF & ABE

Area of triangle $\frac{1}{2} \times DB \times EF$

$$\frac{\text{Ar}(\triangle ADE)}{\text{Ar}(\triangle DBE)} = \frac{\frac{1}{2} \times AD \times EF}{\frac{1}{2} \times DB \times EF} = \frac{AD}{DB} \quad \dots\dots(1)$$

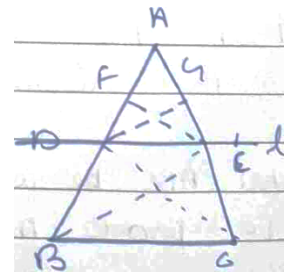
Similarly

$$\frac{\text{Ar}(\triangle ADF)}{\text{Ar}(\triangle DCE)} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG} = \frac{AE}{EC} \quad \dots\dots(2)$$

$$\text{Ar}(\triangle DBE) = \text{Ar}(\triangle DCE) \quad \dots\dots(3)$$

From (1) (2) (3) we have

$$\frac{AD}{DB} = \frac{AE}{EC}$$



<sol_end>

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<page_start no = 5>

<sol_start id=28>

28. Let Sagar age be = x

Let Tiru age be = y

$$2(5-x) = (5-y)$$

$$10-2x = 5-y$$

$$-2x + 5 + y$$

$$-2x + y + 5 = 0 \quad \dots\dots(1)$$

$$-2x + y = -5$$

$$(10 + x) + 10 = (10 + y)$$

$$20 + x = 10 + y$$

$$x - y + 10 = 0 \quad \dots(2)$$

$$x - y = 10$$

comparing (1) & (2)

$$-2x + y + 5 = 0$$

$$x - y + 10 = 0$$

$$-x + 15 = 0$$

$$-x = -15$$

$$x = 15$$

$$15 - y = -10$$

$$-y = -25$$

$$y = 25$$

Age of Sagar = 15

Age of Tiru = 25

<sol_end>

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<page_start no = 6>

<sol_start id=29>

29. A(-5, 7), B(-4, 5), C(-1, -6), D(4, 5)

Join A to C

$$\therefore \text{Ar}(\text{ABCD}) = \text{Ar}(\text{ABC}) + \text{Ar}(\text{ADC})$$

$$\text{Area of } \triangle \text{ABC} = \frac{1}{2} [-5(-5+6) - 4(-6-7) - 1(7+5)]$$

$$= \frac{1}{2} 35 \text{ sq. unit}$$

$$\text{Ar of } \triangle \text{ADC} = \frac{1}{2} [-5(5+6) + (4(-6-7) - 1(7-5) - 1(7-5))]$$

$$= \frac{1}{2} (-109) = [\text{Area can be negative}]$$

$$= \frac{1}{2} 109 \text{ sq. units}$$

$$\text{Ar}(\text{ABD}) = \frac{1}{2} 35 + \frac{1}{2} 109$$

$$= \frac{1}{2} 144$$

272 sq. unit

$$= 72 \text{ sq. unit}$$

<sol_end>

<sol_start id=30>

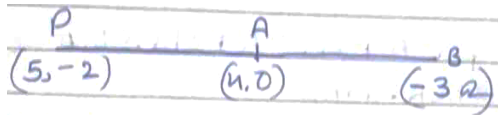
$$30. \quad 31353 = 861 \times 1 + 492$$

$$861 = 492 \times 1 + 369$$

<sol_end>

<sol_start id=31>

31. Let the point $P(x, 0)$ is equidistant from the pts.



$$\text{i.e. } AP = BP$$

$$AP^2 = BP^2$$

$$(5-x)^2 + (-2-0)^2 = (-3-x)^2 + (2-0)^2$$

$$= x^2 + 25 - 10x + 4 = x^2 + 9 + 6x + 4$$

$$= 10x + 6x + 9 + 4 - 4x - 25 = 0$$

$$16 - 16 = 0$$

$$4 = \frac{16}{16} = 1$$

So $(1, 0)$ is the point.

<sol_end>

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<page_start no = 7>

<sol_start id=32>

$$32. \quad \tan A + \sin A = m$$

$$\tan A - \sin A = n$$

$$m^2 = (\tan A + \sin A)^2 = \tan^2 A + \sin^2 A + 2 \tan A \times \sin A$$

$$n^2 = (\tan A - \sin A)^2 = \tan^2 A + \sin^2 A - 2 \tan A \times \sin A$$

$$= m^2 - n^2 = \tan^2 A + \sin^2 A + 2 \tan A \cdot \sin A - (\tan^2 A + \sin^2 A - 2 \tan A \cdot \sin A)$$

$$= 4 \tan A \cdot \sin A$$

$$mn = (\tan A + \sin A) (\tan A - \sin A)$$

$$= \tan^2 A - \sin^2 A = \frac{\sin^2 A}{\cos^2 A} = \sin^2 A$$

$$= \sin^2 A (\sec^2 A - 1)$$

W.K.T.

$$\sec^2 A - \tan^2 A = 1$$

$$= \sin^2 A \times \tan^2 A = \sqrt{mn}$$

$$\text{LHS} = 4\text{RHS}$$

$$m^2 - n^2 = 4\sqrt{mn}$$

<sol_end>

<page_end>

<page_start no = 8>

<sol_start id=33>

33. $x - y + 1 = 0$

$$y = x + 1$$

$$\text{put } x = -1$$

$$y = 0$$

$$\text{put } x = 1$$

$$y = 2$$

$$\text{put } x = 2$$

$$y = 3$$

x	-1	1	2
y	0	2	3

$$3x + 2y - 12 = 0$$

$$2y = (-3x + 12)$$

$$y = \frac{-3x + 12}{2}$$

$$\text{Put } x = 0$$

$$y = 0 + \frac{12}{2} = 6$$

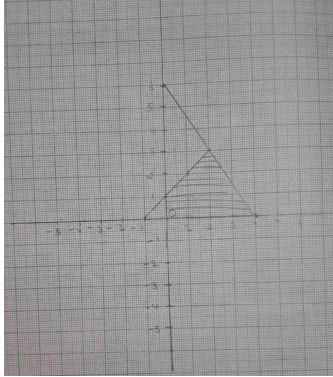
X	0	-2	4
Y	6	3	0

$$x = 2$$

$$y = \frac{-6 + 12}{2} = 3$$

$$x = 4$$

$$y = \frac{-12 + 12}{2} = 0$$



= Area of triangle

$$\frac{1}{2} \times B \times H$$

$$\frac{1}{2} \times 5 \times 3$$

$$= \frac{1}{2} \times 15 = 7.5 \text{ sq. unit}$$

<sol_end>

<sol_start id=34>

Q.34 $p(x) = 2x^3 - 4x - x^2 + 2$

Two zeros are $\sqrt{2}, -\sqrt{2}$

$$\Rightarrow (x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$$

$x^2 - 2$ is the factor of $p(x)$

Apply the division Algorithm

$p(x)$ $2x^3 - 4x - x^2 + 2$ & $x^2 - 2$

remainder $r = 0$

$$p(x) = q(x) \times g(x)$$

$$\therefore 2x^3 - 4x - x^2 + 2 = (x^2 - 2)(2x - 1)$$

$$= (x - \sqrt{2})(x + \sqrt{2})(2x - 1)$$

$$= (x - \sqrt{2})(x + \sqrt{2})(2x - 1) = 0$$

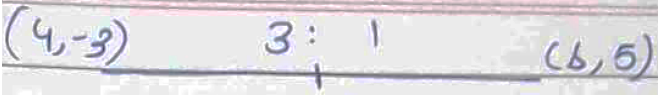
$$= x = \sqrt{2} \quad x^2 - \sqrt{2}, \quad x = \frac{1}{2}$$

<sol_end>

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<sol_start id=35>

35. 

$$P(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$x = \frac{3(8) + 1(4)}{3 + 1} = \frac{24 + 4}{4} = \frac{28}{4} = 7$$

$$y = \frac{3(5) + 1(-3)}{3 + 1} = \frac{15 - 3}{4} = \frac{12}{4} = 3$$

P(7, 3) is the required point.

<sol_end>

<sol_start id=36>

<sub_id = 1>

36. (i) (0, 10)

<sub_id end>

<sub_id = 2>

(ii) (3, 5)

<sub_id end>

<sub_id = 3>

(iii) 4m

<sub_id end>

<sol_end>

<sol_start id=37>

<sub_id = 1>

37. (i) Parabola

<sub_id end>

<sub_id = 2>

(ii) 1

<sub_id end>

<sub_id = 3>

(iii) -3, 2

<sub_id end>

<sol_end>

<sol_start id=38>

<sub_id = 1>

38. (i) 24 cm^2

<sub_id end>

<sub_id = 2>

(ii) RHS

<sub_id end>

<sub_id = 3>

(iii) 16:81

<sub_id end>

<sol_end>

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